

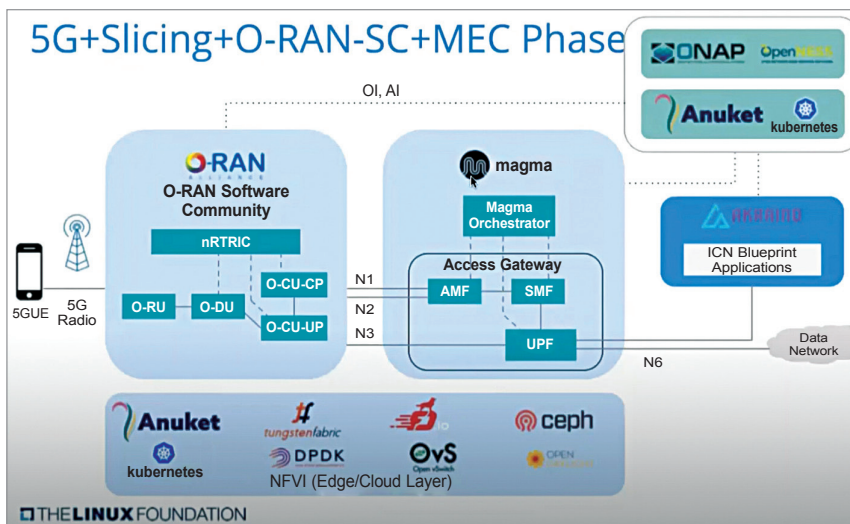


Fig. 4: The third phase of 5G modernisation

The edge project stack is a very important attribute. For example, if a sensor wakes up and dumps data to an Amazon cloud every week, it's an IoT application. It's not an edge application, and vice versa. Hence, the LF edge as an umbrella is unifying the frameworks.

Akraino R4 blueprints

Akraino is a blueprint concept. The open community comes together to define a use case. And then, through open governance, tests out the use case and interoperability all the way from hardware to the application, and then submits the results so that developers can replicate it.

That's a blueprint, a declarative configuration, with CI/CD documentation. The one that has become very popular with telcos is PCEI (public cloud edge interface).

So, the question arises, how the public cloud and the service provider at the edge have a common set of interfaces? There are no standards that exist.

Use cases

ORAN use cases include the slicing and the disaggregation that comes with the open RAM. And then on the Kubernetes, it's the cloud, multi-cloud hybrid layer below with the standard compliance and then ONAP, which is the largest and the most important. It goes through 5G network slicing and all the use cases for telecom. This is really the fundamental platform, 10

plus million lines of code that is at the heart of network automation and network automation is a must for 5G.

So as a global community, ONAP has matured and is one of the most important projects to enable end to end systems.

In case of intermittent connectivity, the user needs an on-prem framework, which is Eve or edge virtualisation. EdgeXFoundry, the open source vendor neutral, is by far the largest framework for IoT. It abstracts southbound and northbound—all the messaging protocols and everything northbound—and this is where building automation, smart process control, water cities happens.

LF networking may now be working on about ten projects—Magma cloud native computing foundation, Kubernetes, or a software community. We are taking these communities and doing an open 5G super blueprint integration.

How it comes together

The 5G super blueprint comes pre-baked and only a small set of integration and support is needed by the community or their vendors. So, instead of multiple vendors doing things in different ways, now you have it done in the community and then the vendor can provide innovation and differentiation on top.

5G will modernise today's energy grid with awareness monitoring across distribution systems.

It will roll out in three phases:

5G core + MEC phase, 5G E2E + MEC phase, and 5G + Slicing + O-RAN-SC + MEC phase.

In the first phase the communities are using the emulators on the RAN side, but Magma is being inserted with ONAP and Anuket on Kubernetes. There are some components from Intel called openness also coming in but, in general, those are the big new components.

The second phase is bringing in a commercial genome piece. The RAN side of things are integrated, and then you run a slice through it. So, the end-to-end network slicing comes in.

The last phase, which is towards the end of the year, is when ORAN comes in. What's already happened is ONAP and ORAN are already deeply integrated with interfaces. There is a lot of component reuse between projects because these are all microservices, containerised modules, and you can just use them as needed and it will just get you a head start. And that's already happening.

Conclusion

Completing the solution stacks to integrate a fully open networking and edge stack, speeding up 'backend interior and deployment process' with open compliance, integrated deployment in vertical markets to start embracing, utilising and building on networking, cloud, edge stacks are seen as the top priorities.

To conclude, blueprints are fairly matured and deployed. "Most of the projects are live in production. The first step is that it needs to happen. In several use cases, the 5G node is now merged from the open-ended interface side of things. And so, Magma itself, as an open mobile packet core in the Q3 timeframe, will be fairly mature.

The testing of these blueprints should be ready by the end of the year. Till then it is a proprietary mobile packet core. Pretty much everything that is at the higher layers of the stack is mature. At the lower layers, radio will stay proprietary, but the disaggregation concept is mature. **EFY**

The article is based on the tech talk by Arpit Joshipura, GM, Networking Edge and IoT, Linux Foundation at Open Source Conference 2021.